JDBC Connection Walkthrough Guide

This document provides a complete, step-by-step procedure to set up a MySQL database, integrate the JDBC driver, and run a Java program to fetch records.

Step 1: Create and Set Up MySQL Database

Open your terminal or MySQL command line utility and run the following commands to create your database structure and populate records:

```
CREATE DATABASE college;
USE college;

CREATE TABLE student (
    id INT,
    name VARCHAR(50)
);

INSERT INTO student VALUES
(101, 'Ramesh'),
(102, 'Haswika'),
(103, 'Harshith');
```

Expected MySQL Output Verification:

Run 'SELECT * FROM student;' to verify your data looks like this:

```
+------+----------+
| id   | name     |
+------+----------+
| 101  | Ramesh   |
| 102  | Haswika  |
| 103  | Harshith |
+------+----------+
3 rows in set (0.00 sec)
```

Step 2: Create the Java Source File

Save the following source code into a file named 'JdbcDemo.java'. Remember to change "password" to your actual MySQL root user password.

```
import java.sql.*;

public class JdbcDemo {
```

```
    public static void main(String[] args) {
        try {
            // Connect to MySQL
            Connection con = DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/college", "root", "password"
            );

            // Create Statement
            Statement stmt = con.createStatement();

            // Execute Query
            ResultSet rs = stmt.executeQuery("SELECT * FROM student");

            // Display Records
            while (rs.next()) {
                System.out.println(rs.getInt("id") + " " +
rs.getString("name"));
            }

            // Close Connection
            con.close();
        } catch (Exception e) {
            System.out.println(e);
        }
    }
}
```

Step 3: Download and Extract MySQL JDBC Driver

Execute the following commands in your Linux/macOS terminal environment to pull the specific connector archive and unpack it:

```
1. wget https://dev.mysql.com/get/Downloads/Connector-J/mysql-connector-j-
9.4.0.tar.gz
2. tar -xzf mysql-connector-j-9.4.0.tar.gz
3. Verify file path structure:
   ls mysql-connector-j-9.4.0/
```

Expected File Extraction Output:

```
INFO_BIN  INFO_SRC  LICENSE  README  mysql-connector-j-9.4.0.jar
```

Step 4: Compile and Run the Application

Compile your Java file by linking the extracted JAR archive to your compilation classpath context:

```
javac -cp .:mysql-connector-j-9.4.0/mysql-connector-j-9.4.0.jar
JdbcDemo.java
```

Run the application package passing down the identical classpath variables:

```
java -cp .:mysql-connector-j-9.4.0/mysql-connector-j-9.4.0.jar JdbcDemo
```

Expected Final Output Execution Screen:

```
101 Ramesh
102 Haswika
103 Harshith
```